Customer Segmentation via Clustering

Generated by uber-polya

2026-02-26

Mode: Full Pipeline

1. Problem Formulation

Problem Type: Prove
Domain: Machine Learning

Universe

- Elbow Analysis: \{k_range, inertias\}
- Kmeans: \{k, labels, centroids, silhouette, inertia\}
- Dbscan: \{eps, labels, n_clusters, n_noise, silhouette\}
- Gmm: \{n_components, labels, bic, aic, silhouette\}

Variables

Symbol	Meaning	Type / Range
x	decision variable	\mathbb{{R}}

Structure

- **Data**: ~500 synthetic customers with 4 purchasing features
- **Features**: annual_spending, purchase_frequency, avg_order_value, days_since_last_purchase
- **Natural segments**: 3 ground-truth clusters with overlapping boundaries
  - **High Value** (~150 customers): high spending, high frequency, low recency
  - **Moderate** (~200 customers): medium spending, medium frequency
  - **At Risk** (~150 customers): low spending, low frequency, high recency
- **Questions**: (1) How many segments exist? (2) Which method recovers them best? (3) What defines each segment?

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) K-Means**: partition-based clustering minimizing within-cluster variance; assumes spherica

(2) DBSCAN**: density-based clustering that finds arbitrarily shaped clusters and identifies noi

(3) GMM (Gaussian Mixture Model)**: probabilistic model assuming data is generated from a m

(4) Silhouette score**: measures how similar an object is to its own cluster vs. the nearest neig

(5) Elbow method**: heuristic for choosing k by finding the "knee" in the inertia curve

(6) Cluster profiles**: per-cluster summary statistics that translate numeric labels into busines

Approach

Suggested approach: K-Means + DBSCAN + GMM

Complexity class:

Available tools: K-Means + DBSCAN + GMM

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	K-Means + DBSCAN + GMM
Time:	0.2149s

Solution Details

Method: 1. StandardScaler: Zero-mean, unit-variance normalization of all features. Essential for distance-based methods (K-Means, DBSCAN) where feature scales differ by orders of magnitude.

2. Elbow method: Run K-Means for k=2..8, plot inertia (within-cluster sum of squares). The "elbow" where marginal improvement drops suggests the natural number of clusters.

3. K-Means (k=3): Lloyd's algorithm with k-means++ initialization. Fast, deterministic given seed, assumes spherical clusters. Reports centroids, labels, inertia, and silhouette score.

4. DBSCAN: Density-based clustering that discovers clusters of arbitrary shape. The eps parameter is set via k-distance heuristic (knee of the sorted 5-nearest-neighbor distance curve). Reports labels, number of clusters, number of noise points, and silhouette score.

5. GMM (3 components): Gaussian Mixture Model fit via Expectation-Maximization. Soft clustering that models each cluster as a multivariate Gaussian. Reports labels (hard assignment), BIC, AIC, and silhouette score.

6. Comparison: Methods ranked by silhouette score; best method selected.

7. Profiling: For the best method, compute mean feature values per cluster to build interpretable segment descriptions.

Verification

✓	Kmeans Silhouette Gt 04	Passed
✓	Kmeans Found 3 Clusters	Passed
✓	Dbscan Found 2 To 5 Clusters	Passed
✓	Gmm Bic Finite	Passed
✓	Cluster Sizes Reasonable	Passed
✓	Profiles Distinguishable	Passed
✓	All Passed	Passed

3. Interpretation

The Question

- **Data**: ~500 synthetic customers with 4 purchasing features
- **Features**: annual_spending, purchase_frequency, avg_order_value, days_since_last_purchase
- **Natural segments**: 3 ground-truth clusters with overlapping boundaries
  - **High Value** (~150 customers): high spending, high frequency, low recency
  - **Moderate** (~200 customers): medium spending, medium frequency
  - **At Risk** (~150 customers): low spending, low frequency, high recency
- **Questions**: (1) How many segments exist? (2) Which method recovers them best? (3) What defines each segment?.

The Answer

A feasible solution was found.

What This Means

- Elbow analysis: inertia curve for k=2..8 shows clear elbow at k=3
- K-Means (k=3): silhouette score > 0.4, recovers 3 well-separated clusters
- DBSCAN: finds 2--5 clusters depending on eps (determined via k-distance heuristic)
- GMM (3 components): finite BIC/AIC, comparable silhouette to K-Means
- Best method: selected by highest silhouette score
- Cluster profiles: distinct mean feature values per cluster (distinguishable in standardized space)
- All 7 verification checks pass

1. StandardScaler: Zero-mean, unit-variance normalization of all features. Essential for distance-based methods (K-Means, DBSCAN) where feature scales differ by orders of magnitude.

2. Elbow method: Run K-Means for k=2..8, plot inertia (within-cluster sum of squares). The "elbow" where marginal improvement drops suggests the natural number of clusters.

3. K-Means (k=3): Lloyd's algorithm with k-means++ initialization. Fast, deterministic given seed, assumes spherical clusters. Reports centroids, labels, inertia, and

silhouette score.

4. DBSCAN: Density-based clustering that discovers clusters of arbitrary shape. The eps parameter is set via k-distance heuristic (knee of the sorted 5-nearest-neighbor distance curve). Reports labels, number of clusters, number of noise points, and silhouette score.

5. GMM (3 components): Gaussian Mixture Model fit via Expectation-Maximization. Soft clustering that models each cluster as a multivariate Gaussian. Reports labels (hard assignment), BIC, AIC, and silhouette score.

6. Comparison: Methods ranked by silhouette score; best method selected.

7. Profiling: For the best method, compute mean feature values per cluster to build interpretable segment descriptions.

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio